

MUHAMMAD BIN ALI (Gen AI Engineer)

Balapur, Hyderabad, Telangana, 500005

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PROFESSIONAL SUMMARY

- Data Scientist with over 1 year of experience in machine learning, deep learning, and natural language processing (NLP). Proven ability to deliver high-performing models and intuitive solutions, achieving up to 95% accuracy in various applications. Adept at leveraging cutting-edge tools and frameworks for real-world challenges, with a focus on optimizing performance and user engagement.
- Transformed complex datasets into actionable insights, improving model accuracy by over 20% through advanced statistical analysis, feature engineering, and optimization techniques.
- Leveraged cutting-edge AI frameworks like TensorFlow, Keras, and Hugging Face to build solutions for over 5 diverse applications, including NLP and Generative AI.
- Collaborative team player with excellent communication skills, able to explain complex AI concepts to both technical and non-technical stakeholders, fostering informed decision-making and driving project success.
- Passionate about using data and machine learning to drive business growth and innovation, constantly staying at the forefront of AI advancements to deliver scalable, high-impact solutions. With a strong problem-solving mindset and collaborative approach, I am committed to turning complex challenges into opportunities for success.

SKILLS

- **Languages :** Python, SQL
- **Python Libraries :** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, NLTK, Gensim, Flask, Streamlit, LangChain
- **Machine Learning :** Regression, Classification, Clustering, Dimensionality Reduction, Ensemble Methods (Random Forest), Time Series, Recommendation System
- **Deep Learning :** Neural Networks (ANN, CNN, RNN, LSTM), Autoencoders, Model Optimization, Hyperparameter Tuning, Frameworks (TensorFlow, Keras)
- **Tableau :** Data Visualization, Dashboard Creation, Report Generation, Data Blending, Data Integration, Calculations (LOD, Table Calculations), Parameters, Filters, Aggregations, Trend Analysis, Time Series Analysis, Advanced Visualizations (Heatmaps, Tree Maps, Waterfall)
- **Natural Language Processing (NLP) :** Text Preprocessing (Tokenization, Lemmatization, Stemming), Bag of Words, TF-IDF, Word Embeddings (Word2Vec, AvgWord2Vec), Named Entity Recognition (NER), Sentiment Analysis, Text Classification, Text Generation, Transformers , Sequence-to-Sequence Models, Attention Mechanism, Question Answering Systems, Text Summarization, NLTK, Hugging Face Transformers
- **Generative AI :** Text Generation (GPT, GPT-3, GPT-4), Language Modeling, Transformers, Transfer Learning, Zero-shot Learning, Prompt Engineering, Text-to-Speech, Speech-to-Text, OpenAI API, Hugging Face Transformers
- **Databases & Tools :** Microsoft SQL Server, MySQL, PostgreSQL, Hadoop, Git, GitHub, Pinecone, Chroma DB, Cassandra DB
- **Cloud Technologies :** AWS (EC2) , Google Cloud Platform (GCP), Microsoft Azure, Heroku
- **Data Visualization Tools :** Tableau, Matplotlib, Seaborn
- **Statistics:** Descriptive Statistics, Inferential Statistics, Hypothesis Testing (A/B Testing), Confidence Intervals, p-Values, Regression Analysis (Linear, Logistic, Multivariate), ANOVA (Analysis of Variance), Time Series Analysis, Probability Distributions (Normal, Binomial, Poisson), Statistical Modeling, Experimental Design, Sampling Techniques, Statistical Significance, Correlation, Chi-Square Test

EXPERIENCE

Data Science Intern : NullClass

Sep 2024 to Jan 2025

Hyderabad, Telangana

- **Emotion Detection from Facial Images** :- Developed and trained a Convolutional Neural Network (CNN) for image classification using TensorFlow and Keras. Utilized ImageDataGenerator for data augmentation and preprocessing of grayscale images.
- **Age and Gender Detection from Facial Images** :- Developed an advanced graphical user interface to classify the age and gender of a person based on a photograph using OpenCV and Keras. Also, I have a graphical interface to classify a person's age and gender using OpenCV and Keras.

PROJECTS

NLP Text Classifier | Python, Bidirectional LSTM and TensorFlow

March 2024

- Developed an **NLP** pipeline to classify text data into binary categories with an accuracy of 90.12%, leveraging advanced preprocessing techniques like stemming and stopwords removal.
- Evaluated the model using confusion matrix metrics, achieving 1,852 correct predictions out of 2,055 test samples, showcasing its reliability in real-world applications.
- Integrated feature extraction methods like TF-IDF and word embeddings (Word2Vec, AvgWord2Vec), resulting in a 15% improvement in model performance, enabling the classifier to process and understand complex textual patterns more effectively.

Weather Information Assistant | Python, Open AI (LLM) GPT-3.5 and Rapid API

April 2024

- Built a conversational assistant using Open AI's GPT-3.5 to deliver real-time weather insights, Achieved an accuracy of 95% in temperature predictions.
- Managed over 1,000 queries monthly, with automated weather data retrieval for 500+ unique locations, optimizing response times by 30% with efficient API integration.
- Integrated a Weather API via Rapid API to fetch current weather data based on user queries.
- Designed a function to retrieve and display current weather information for specified locations.

Q&A Chatbot | Python, Google Gemini (LLM) and Streamlit

May 2024

- Developed an advanced Q&A chatbot using **Google's Gemini model**, generating 95% accurate responses to both text and image-based queries.
- Implemented the chatbot interface using **Streamlit**, allowing users to upload images and input text prompts for analysis.
- Enhanced user engagement by 40%, handling over 1,500 interactions monthly, with a user-friendly interface that streamlined user input, improving session durations by 25%.

Health Management App | Python, Google Gemini (LLM) and Streamlit

June 2024

- Developed and deployed a comprehensive health management system leveraging Google Generative AI (Gemini-pro-vision model) and Streamlit, enabling real-time calorie estimation from food images with 95% accuracy, while ensuring secure API integration and seamless user experience.
- Utilized **Google Generative AI** with the **Gemini-pro-vision** model to analyze food images and calculate calorie intake.
- Implemented a function to generate detailed nutritional responses based on user input and uploaded images.
- Created a user-friendly interface with **Streamlit**, enabling users to input text prompts and upload images for calorie estimation.

Blog Generation App | Python, Llama 2 (LLM) Model and Streamlit

July 2024

- Leveraged the **Llama 2** model with **Transformers** for generating content based on user inputs.
- Achieved a 98% satisfaction rate in content relevance and quality, with optimized prompt handling that decreased response time by 20%, driving 50% repeat usage from content creators.
- Created a dynamic web interface with Streamlit, allowing users to input blog topics, word count, and target audience.
- Built a function to format prompts and obtain responses from the **Llama 2** model, enhancing user interaction and content generation capabilities.

EDUCATION

ISL Engineering College, Osmania University

Bachelor of Engineering (BE) in Computer Science Engineering (CSE)

Hyderabad, Telangana

2019 to 2023

CERTIFICATIONS

- Python For Data Science - IBM
- Data Science Master's Program - Simplilearn
- Python For Data Science - Simplilearn
- Machine Learning - Simplilearn
- Deep Learning - Simplilearn
- Tableau - Simplilearn

CLOUD DEPLOYMENT and CONTAINERS

Cloud Platform Experience : Proficient in deploying applications on various cloud platforms.

- **AWS EC2, Azure Deployment, Heroku, Google Cloud Platform (GCP)**

TOOLS

ChatGPT, DeepSeek, Google-Gemini, Microsoft Bing, Windows 10 and 11, MS Office (Word, PowerPoint)

- Excellent knowledge in utilizing various tools such as **ChatGPT** and **Google-Gemini** for AI-driven applications.

SOFT SKILLS

Soft Skills : Demonstrated excellence in various soft skills critical for professional success.

- **Communication Skills**: Strong verbal and written communication skills.
- **Teamwork** : Proven ability to work effectively in a team environment.
- **Problem-Solving** : Excellent problem-solving and analytical skills.
- **Leadership** : Leadership experience in managing projects and guiding team members.
- **Time Management** : Strong organizational and time management abilities.
- **Adaptability** : Highly adaptable to changing environments and demands.

DECLARATION

Declaration : I hereby declare that the above information is true to the best of my knowledge and belief.

- Name : Muhammad Bin Ali
- Date : February 6, 2025
- Place : Balapur, Hyderabad, Telangana.